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KGiSL Campus, Saravanampatti, Coimbatore – 641 035

JAVA PROJECT REPORT ON

ONLINE EVENT TICKET BOOKING AND RESERVATION SYSTEM

Submitted by
RAMYA R
25BIT065

Under the Guidance of
DHILEBAN A
Technical Trainer
Technical Department, KG College of Arts and Science.

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DECLARATION

I hereby declare that the project entitled **“Online Event Ticket Booking and Reservation System”** submitted in partial fulfillment of the requirements for the course Java Programming is a record of my work carried out under the guidance of Dhileebana, Technical Trainer, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title at this or any other institution. Sources of information used in preparing this report have been acknowledged wherever applicable.

April 2026
Coimbatore

RAMYA R
25BIT065

CERTIFICATE

This is to certify that the project entitled **“Online Event Ticket Booking and Reservation System”** is a bona fide work carried out by **Ramya R (25BIT065)**, B.Sc. Information Technology, KG College of Arts and Science, in partial fulfillment of the requirements for the course Java Programming.

The project has been completed under the supervision and guidance of the Technical Department during the academic period 2025–2026.

Place: Coimbatore
Date: _____

Guide	Head of the Department
Dhileebana	_____
Technical Trainer, KGCAS	Technical Head, KGCAS

Submitted for the Viva-Voce Examination held on _____

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ABSTRACT

The Online Event Ticket Booking and Reservation System is a Java-based application designed to make event ticket booking simple, fast, and organized. Users can view available events, check ticket prices and seat availability, select an event, enter the number of tickets, and receive a booking confirmation. The sample application demonstrates event listings such as Music Concert, Movie Show, and Tech Seminar.

The project uses Core Java concepts including classes, objects, methods, arrays or collections, conditional statements, loops, and input handling. It reduces manual calculation and helps maintain available-seat counts. This basic version is menu-driven and stores information during program execution; database connectivity and online payment can be added in a future version.

1. INTRODUCTION

Events such as music concerts, movie shows, seminars, workshops, and college programmes attract many participants. Managing reservations manually can be time-consuming and may lead to duplicate bookings, calculation mistakes, and confusion about available seats.

The Online Event Ticket Booking and Reservation System provides a simple computerized method for displaying event information and processing ticket reservations. The user can select an event, specify ticket quantity, calculate the total cost, and view booking details. The system updates the remaining seat count after a successful reservation.

1.1 Objective

• To develop a Java-based event ticket booking application.

• To display event names, ticket prices, and seat availability.

• To allow users to choose an event and enter ticket quantity.

• To calculate the booking amount accurately.

• To update the available seats after a successful booking.

• To demonstrate object-oriented programming concepts in a practical project.

1.2 OVERVIEW

The application presents a menu of available events. Each event contains an event ID, name, ticket price, and number of available seats. A user selects an event and enters the required number of tickets. The program validates the selection and checks whether enough seats are available.

If the booking is valid, the system calculates the total amount by multiplying the ticket price by the number of tickets. It then reduces the available-seat count and displays a confirmation containing the event name, quantity, and total amount. If the selection is invalid or the requested seats exceed availability, an error message is shown.

1.3 FEATURES

• Event Listing – displays available events and their ticket prices.

• Seat Availability – shows the number of seats remaining.

• Event Selection – allows the user to select an event using its ID or menu number.

• Quantity Validation – checks that the ticket quantity is positive.

• Booking Calculation – calculates the total ticket cost.

• Reservation Confirmation – displays booking details after successful processing.

• Seat Update – decreases available seats after booking.

• Exit Option – terminates the program safely.

PROJECT BENEFITS AND APPLICATIONS

Benefits

• Saves time compared with manual ticket reservation.

• Reduces arithmetic errors in total-price calculation.

• Provides clear information about ticket prices and seat availability.

• Helps avoid bookings that exceed the available seat count.

• Provides a useful learning example of Java methods and classes.

Applications

• College cultural events and symposiums.

• Music concerts and stage shows.

• Movie or theatre ticket reservations.

• Workshops, seminars, and technical conferences.

• Small event-organizing teams requiring a basic reservation tool.

Technologies Used

Technology	Purpose
Java	Main programming language
Core Java / OOP	Classes, objects, methods, conditions and loops
Scanner	Reads keyboard input
Array / ArrayList	Stores event details in memory
Console	Displays the menu, booking details and messages

2. PROGRAM ANALYSIS

Program analysis identifies the problem, the requirements, and the expected behaviour before implementation. For this project, the main task is to provide a structured way to browse events and reserve tickets while maintaining correct seat counts.

2.1 Problem Definition

When ticket reservations are handled manually, organizers may need to check paper lists, calculate prices by hand, and repeatedly update seat availability. These steps can result in slow service, duplicate entries, incorrect totals, and overbooking.

2.2 Proposed Solution

• Maintain a list of events with IDs, names, prices and seat counts.

• Display event information in a readable menu.

• Accept the user’s event selection and ticket quantity.

• Validate input and available seats before confirming the booking.

• Calculate the total price automatically.

• Update seat availability and show a confirmation message.

2.3 SCOPE AND LIMITATIONS

Scope

• List predefined events and their ticket prices.

• Check event selection and seat availability.

• Reserve one or more tickets in a single booking.

• Calculate and display the total cost.

• Update the in-memory seat count after a confirmed reservation.

• Use the project as a foundation for a larger ticketing application.

Limitations

• The basic version stores data in memory, so records may be lost when the program closes.

• It does not provide a live website or mobile application interface.

• No database is connected in the basic implementation.

• Online payment, email notifications, QR tickets and refunds are not included.

• Multi-user concurrency is not supported; simultaneous bookings would require server-side validation.

3. SYSTEM DESIGN

System design describes the components of the application and how information moves between them. The project follows a menu-driven structure. The user interacts with the menu, the event module supplies event details, the booking module validates the reservation, and the confirmation module displays the result.

3.1 Main Modules

Module	Description
Event Display	Shows event ID, event name, price and available seats.
Event Selection	Reads and validates the selected event.
Seat Validation	Checks ticket quantity and remaining seats.
Booking Calculation	Calculates total = ticket price × ticket quantity.
Reservation Processing	Confirms a valid reservation and updates seats.
Confirmation Display	Shows event name, ticket count and amount.
Exit	Ends program execution.

3.2 SYSTEM WORKFLOW

The following sequence describes the normal booking process:

```
START
  |
Display event list and prices
  |
Read event selection
  |
Is selection valid? ---- No ----> Display error
  | Yes
Read ticket quantity
  |
Are quantity and seats valid? -- No --> Display message
  | Yes
Calculate total price
  |
Reduce available seats
  |
Display booking confirmation
  |
Return to menu or Exit
  |
 END
```

Data Design

An Event object can contain eventId, eventName, ticketPrice, and availableSeats. A booking operation uses the selected Event object and requested quantity to calculate the amount and update the seat count. Keeping these values together makes the program easier to maintain.

4. METHODOLOGY

The project is developed in stages so that requirements, design, implementation, and testing can be handled systematically.

4.1 Problem Analysis

The requirements were identified as event display, event selection, ticket quantity input, price calculation, seat validation, reservation confirmation, and safe exit.

4.2 System Design

The application was divided into event display, validation, booking calculation, reservation processing, and output modules. A menu-driven console interface was selected to keep the project simple and easy to demonstrate.

4.3 Program Development

• Create an Event class to store event details.

• Initialize sample events such as Music Concert, Movie Show, and Tech Seminar.

• Display the menu using a loop.

• Use conditional statements or switch-case to process the user’s choice.

• Validate ticket quantity and available seats.

• Calculate the total amount and update the seat count.

• Display a clear booking confirmation.

4.4 TESTING AND VALIDATION

Test Case	Expected Result
Display event menu	All predefined events and prices are shown.
Select valid event	Selected event details are accepted.
Enter positive quantity within availability	Booking total is calculated.
Request more seats than available	Booking is rejected with a message.
Enter zero or negative quantity	Invalid quantity message is displayed.
Enter an invalid event option	Invalid selection message is displayed.
Complete valid booking	Available seats are reduced correctly.

4.5 Result and Documentation

The expected result is a working menu-driven program that lists events, validates booking requests, calculates ticket totals, updates available seats, and displays reservation confirmation. Actual execution should be tested in the Java environment and screenshots should be added to the final report.

4.6 ALGORITHM

1. Start the program.

2. Initialize the list of events with IDs, names, ticket prices and seat availability.

3. Display the main menu.

4. Read the user’s event selection.

5. If the user chooses Exit, stop the program.

6. Validate the selected event. If invalid, display an error and return to the menu.

7. Read the required number of tickets.

8. Check that the quantity is positive and does not exceed available seats.

9. Calculate the total amount as ticket price multiplied by quantity.

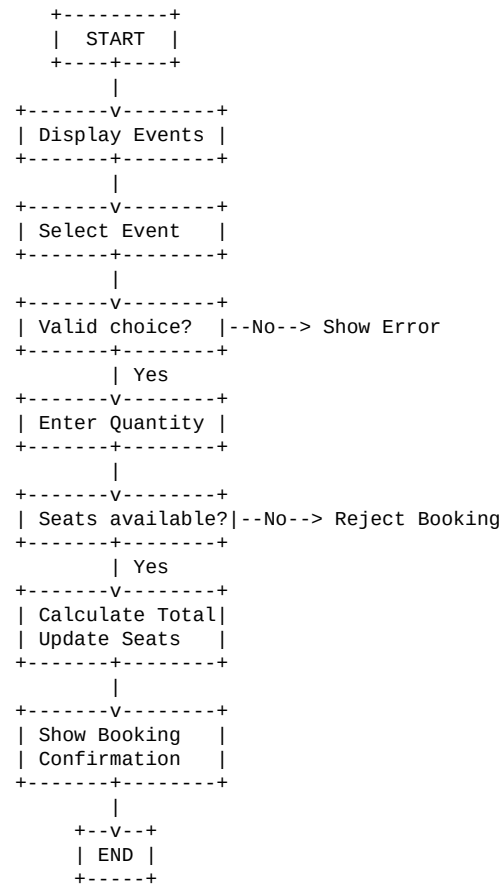
10. Reduce the available-seat count by the number of tickets booked.

11. Display the reservation confirmation and total amount.

12. Return to the main menu and repeat until Exit is selected.

13. End.

4.7 FLOWCHART



5. SOURCE CODE

The following sample Core Java program demonstrates event listing, ticket booking, validation, total calculation, and seat updates. It can be saved as EventBooking.java.

```
import java.util.Scanner;

class Event {
    int id;
    String name;
    double price;
    int seats;

    Event(int id, String name, double price, int seats) {
        this.id = id;
        this.name = name;
        this.price = price;
        this.seats = seats;
    }
}

public class EventBooking {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        Event[] events = {
            new Event(1, "Music Concert", 500.0, 50),
            new Event(2, "Movie Show", 200.0, 100),
            new Event(3, "Tech Seminar", 300.0, 70)
        };

        while (true) {
            System.out.println("\nONLINE EVENT TICKET BOOKING");
            for (Event e : events) {
                System.out.println(e.id + ". " + e.name
                    + " | Rs." + e.price
                    + " | Seats: " + e.seats);
            }
            System.out.println("0. Exit");
            System.out.print("Choose event ID: ");
            int choice = sc.nextInt();

            if (choice == 0) break;
            if (choice < 1 || choice > events.length) {
                System.out.println("Invalid event selection.");
                continue;
            }

            Event selected = events[choice - 1];
            System.out.print("Number of tickets: ");
            int qty = sc.nextInt();

            if (qty <= 0) {
                System.out.println("Enter a positive quantity.");
            } else if (qty > selected.seats) {
                System.out.println("Not enough seats available.");
            } else {
                double total = selected.price * qty;
                selected.seats -= qty;
                System.out.println("\nBOOKING CONFIRMED");
                System.out.println("Event: " + selected.name);
                System.out.println("Tickets: " + qty);
                System.out.printf("Total: Rs.%.2f%n", total);
                System.out.println("Seats remaining: "
                    + selected.seats);
            }
        }
        System.out.println("Thank you for using the system.");
        sc.close();
    }
}
```

6. INPUT AND OUTPUT SCREENS

The following is an illustrative console output for the sample program. Replace or supplement this sample with screenshots captured from the actual Java program during execution.

Figure 1: Event Menu

```
ONLINE EVENT TICKET BOOKING
1. Music Concert | Rs.500.0 | Seats: 50
2. Movie Show | Rs.200.0 | Seats: 100
3. Tech Seminar | Rs.300.0 | Seats: 70
0. Exit
Choose event ID: 1
Number of tickets: 2

BOOKING CONFIRMED
Event: Music Concert
Tickets: 2
Total: Rs.1000.00
Seats remaining: 48
Thank you for using the system.
```

Figure 2: Validation Example

```
Choose event ID: 2
Number of tickets: 150
Not enough seats available.
```

Figure 3: Invalid Quantity

```
Choose event ID: 3
Number of tickets: 0
Enter a positive quantity.
```

7. DISCUSSION AND CONCLUSION

7.1 Discussion

The Online Event Ticket Booking and Reservation System demonstrates how Java can be used to organize a simple ticket reservation process. The event list keeps ticket prices and available seats together. Input validation prevents invalid event selections, zero or negative ticket quantities, and requests that exceed the available seats.

The booking calculation uses the selected event's ticket price and requested quantity. After a valid booking, the program updates the seat count and displays confirmation details. The project applies object-oriented programming through the Event class and uses arrays, loops, conditions, methods and Scanner input.

7.2 Conclusion

The project provides a basic and understandable solution for event ticket booking and reservation. It reduces manual calculations and helps organizers track seat availability within a single program session. It also provides practical experience in applying Core Java concepts to a real-world scenario.

Future improvements may include database connectivity using JDBC, a graphical user interface using Java Swing or JavaFX, user login, online payment integration, email confirmation, QR-code tickets, cancellation and refund management, and a web-based interface. A production system would also require secure storage and server-side controls for simultaneous bookings.

8. REFERENCES

Books

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Project Note

This report describes a basic educational implementation of the Online Event Ticket Booking and Reservation System. Event names, ticket prices, seat counts, and sample outputs are illustrative values used to demonstrate the program logic.